Section 2.1 — Monomials and Polynomials

Math 103: College Algebra

1 Exponents

For each expression below:

a. Identify the base and the exponent.

b. Rewrite as repeated multiplication.

c. Evaluate by multiplying, then check your answer using the exponent key on your calculator.

1. 2^3 2. 3^4 3. $(-4)^2$ 4. 5^3 5. $(-1)^5$

For each of the expressions below:

a. Rewrite using a positive exponent.

b. Evaluate using your rewritten expression, then check your answer using your calculator. Answer as a fraction.

6. 3^{-2} 7. 5^{-1} 8. $\left(\frac{2}{3}\right)^{-3}$ 9. $(-4)^{-3}$ 10. $\left(-\frac{1}{3}\right)^4$

2 Simplifying Monomials

Simplify each monomial below, applying exponent rules as appropriate.

11. b^2b^3 12. $2y^2 \cdot 5y^3z$ 13. $\left(3x^2\right)^3$ 14. $\left(-\frac{3}{2}x\right)^4$

15. $\left(2x^3\right)^4(2xy)^2$ 16. $\frac{3}{8}x^2y \cdot \frac{4}{9}xy^2$ 17. $6hk \cdot 7j^5k \cdot (-2)h^2y$

3 Using Polynomial Vocabulary

For each expression, decide whether it is

A. not a polynomial B. a monomial C. a binomial

D. a trinomial E. a polynomial with 4 or more terms

18. $b+2$ 19. $-\dfrac{2}{3}tu$ 20. x^3 21. $x^2-7xy-y^2$

22. 13 23. $-0.6xy^2$ 24. $\dfrac{4m^2}{n}$ 25. $4m-\dfrac{3}{n^2}$

26. $\frac{5}{8}ab$ 27. $-0.9n^2p^5$ 28. $7r^3-9r^2+r-6$ 29. $\dfrac{5}{x^2}$

30. $5.7x^4y$ 31. y^4 32. x^2+y^{-3} 33. $2x+1$

For each polynomial below:

a. Identify the leading term and leading coefficient, if any.

b. Give the degree of the polynomial.

c. Identify any that are constant, linear, quadratic, or cubic.

34. $6x+8y+3z$ 35. $17.3x^2-3x+2.1$ 36. $18xy^3$ 37. x^3-4x^2+4x+4

38. $7k+5$ 39. 36 40. $25-x^2$ 41. $2u^2+5u^3-u^7$

Give an example of each of the following.

42. A quadratic trinomial in one variable.

43. A homogenous cubic polynomial with four terms. (Use x and y as your variables.)

44. A linear binomial in one variable.

45. A cubic binomial in one variable.

4 Polynomial Functions

Let $f(n) = n^2 - 8n - 3$. Find:

46. $f(2)$
47. $f(-3)$
48. $f(\frac{1}{2})$
49. $f(0)$

Let $g(j) = 22j^3 + 5j^2 - 16j^4 + 21 - 14j$. Find:

50. $g(1)$
51. $g(-2)$
52. $g(\frac{1}{4})$
53. $g(-1)$